Product Adoption & Value Realization

Is what customers bought actually being used —
and what is the gap worth?

Suraj Iyer

INTRODUCTION

Three themes define my career

Senior Product Manager, Microsoft 365 Data Platform. Ten years building data and ML platform products.

Mu Sigma → Bing Ads → Azure Messaging → Microsoft 365. The measurement layer other teams build on.

Customer Centricity

Platform Thinking

Learning

"Building large platforms and data products that solve customers' problems, while I keep learning, is what I love — and that's what brings me here today."

CONTEXT & PROBLEM

Three questions a software business answers precisely. The fourth is the hard one.

What do we sell?
Every SKU, every feature, catalogued ✓

How much did we sell?
To the dollar, to the unit ✓

Who did we sell it to?
Segment, region, industry, contract terms ✓

What are they doing with it?
No shared definition, no shared number ✗

The fourth question is the only one that tells you whether the first three will still be true next year.

Revenue tells you what a customer bought.
It doesn't tell you whether it mattered.

More than half of what customers buy never converts into use — and nobody sees it until the renewal call.

By then the only lever left is a discount. I'll show you exactly how I get to that number — and you should hold me to it.

Value Realization Rate (VRR) is occupancy

VRR = % of their features they use × % of their capacity they use

Everything Acme could get from this SKU is a square. They occupy 26% of it. Acme Financial · $100k/year · 4 features · 10M credits · monthly

Features they use	1 of 4, value-weighted	33%
Capacity they use	667k of 833k credits	80%
Share of the square	33% × 80%	26%

They bought a platform and deployed a point tool. Neither number catches that alone.

100% is not the target.

An account at 100% has no headroom left — they are under-provisioned, not healthy. It is an upsell signal.

It is not a grade.

A low score is unclaimed value you have already sold. The empty part of the square is the opportunity.

THE FRAMEWORK

The same score, four different problems

A single number tells you an account is at 26%. The two factors tell you which of four conversations to have — and who should have it.

Not started — 12 accounts · $3.9M
Deployment rescue · CS leadership

Running hot on one thing — 7 accounts · $1.5M
They will hit the cap and renegotiate on price · Sales + Product

Set up, never scaled — thin at this threshold
Enablement — why hasn't volume grown? · Customer Success

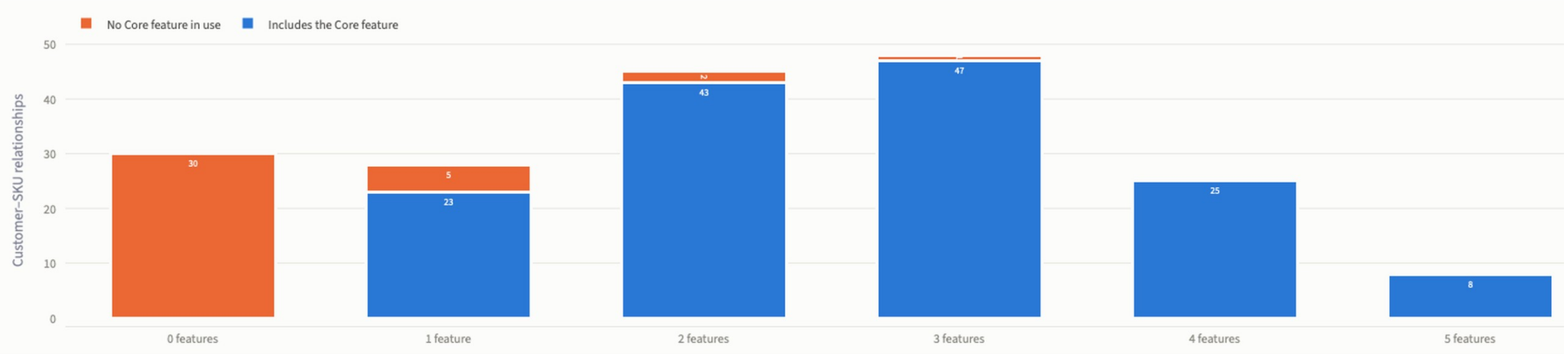
Realizing value — 64 accounts · $7.8M
Expansion conversation · Sales

Bottom-left is a rescue. Bottom-right is a pricing conversation waiting to happen. Top-left is an enablement gap. Top-right is where expansion comes from.

Live in the dashboard, and the threshold is a slider — so “high” is a dial, not a decree. Bubble size is contract value; hovering gives the account, its score, and what it is worth.

EVALUATION & RESULTS

What it found — and the number that should worry you



4% Fully deployed on the widest SKU (with 5 features)

32% use one feature or none

21% use no Core feature at all

Blue means the Core feature is among those in use; orange means it is not. Counted per customer–SKU relationship, not per customer.

LIVE

Four moves in the dashboard

Everything here is read from the pipeline. Nothing is computed in the app — the dashboard and the tests are looking at the same numbers.

1. **Open on the Overview** — The trust line first: data complete through July, three internal accounts excluded, eighty-four duplicate rows removed.

2. **Move the threshold slider** — Every account, plotted. Drag 50% to 70% and watch them redistribute. “High” is a dial, not a decree.

3. **Drill into one account** — Thirteen named features they are paying for and not using — three of them the reason they bought it. This is what a CSM opens.

4. **Open Data Quality** — Every row this pipeline dropped, with a reason and a count, appended on every run.

Move 3 is the one that matters — it is where a metric becomes a phone call.

EVALUATION & RESULTS

One metric, three altitudes, one accountable owner

CPO & GMs

Where to invest, what to repackage

Portfolio score and dollars not converting, cut by segment

Product

Which SKU is failing, and in which of the four ways

Feature-level adoption inside each SKU

CSM & Sales

Which account to call this week

The exact unused features and what they are worth

Every flag ends in a worklist, not a score

What a CSM actually opens:

"Acme Financial is paying for 13 features they did not use this month — **3 of them are the reason the SKU was bought**. Roughly $1.8M a year of contract value sits behind this list."

Then the named features, ranked by how much they matter — Core first.

Not "this account scores 26%."

And it becomes accountable.

Value at Risk = contract value × (1 − VRR). It is in dollars, so it sums by account owner. Quarter over quarter you can see how much at-risk value each owner moved — the bridge from a metric to a motion.

THE ASK

Align on Value Realization Rate as how we measure adoption.

One definition. One calculation. Every team reading the same number.

- **Now** — Agree the metric and the calculation. Everything downstream depends on one number we all accept.
- **Next quarter** — Backtest against real renewal history. Does it predict churn earlier than NRR?
- **After that** — Derive feature weights from outcomes instead of judgment — then, and only then, the incentive conversation.

APPENDIX

Technical details

Architecture and trade-offs · the twelve messy realities · how I know the number is right · what I got wrong

The recommendation ends on the previous slide. Everything past here exists to be challenged.

APPENDIX · ARCHITECTURE

How the data flows

1
Generate
Python + Faker
100 customers · 500 SKUs
2,000 features · 15 months

→

2
Land
BigQuery Sandbox
7 raw tables
row-count verified on load

→

3
Model
15 SQL models
staging → intermediate → marts
full-refresh CTAS

→

4
Serve
5 mart tables
customer · SKU · feature
owner · executive

→

5
Show
Streamlit + Plotly
5 views, read-only
nothing computed in the UI

Four decisions

Full-refresh CTAS, not incremental

The sandbox blocks DML — I verified that with a smoke test rather than reading it in docs. It is also the right call at this volume: idempotent, no state drift.

Ratios diagnose, dollars prioritize

Percentages cannot be summed across customers. Dollars can, so Value at Risk aggregates to any cut of the business.

Streamlit, not React

A deliberate speed trade-off. The deliverable is a decision, not a product.

Nothing hardcoded

Project, dataset, and thresholds all come from config. The pipeline runs against any BigQuery instance by changing one file.

APPENDIX · DATA QUALITY

The hard part: the data is a liar

Twelve realities injected on purpose. Five were specified. Seven I added, because they are what actually breaks adoption metrics in production.

CUSTOMER BEHAVIOUR · 5, SPECIFIED

Shelfware · burn-and-stop · chronic overage · mid-year expansion · brand-new deployments

DATA INTEGRITY · 7, ADDED

Duplicate rows · unlimited contracts · usage on unbought features · lapsed renewals · late-arriving data · internal test accounts · multi-currency

Three of them changed the framework, not just the cleaning rules

Some contracts have no denominator

"Unlimited" deals are scored on feature coverage alone. Substituting 100% would make them look like the healthiest accounts in the portfolio.

There is a third state

A lapsed renewal is not healthy and not at risk — it is not applicable, and produces no row at all. Most frameworks only have two states.

A number can be correct but not final

The current month is marked incomplete and drawn greyed. Hide that, and the first time an exec notices it move, they stop trusting every number.

APPENDIX · VALIDATION

The tests check the answer, not the plumbing

I planted twelve problems in the data, then tested whether the metric found them — without ever letting the pipeline see the labels.

Ground truth the pipeline cannot see

The generator secretly labels each customer's intended behaviour in a column no model reads. The tests ask whether the metric independently rediscovered what was planted.

40 automated tests

Structure, injection, business-answer correctness, and invariants — including one that simply asserts the portfolio value is plausible.

Every dropped row is on the record

Each cleaning step logs rows in, rows out, and why. Appended per run, so a number that moves can be traced to the step that moved it.

What the last run actually did

84	duplicate contract-months removed
3	internal test accounts excluded
395	usage rows for features never bought
684	months with a contract and no usage — counted as zero, not unknown
144	rows in the current month marked still loading
4	months with no active contract — not scored, not shelfware

Every row, every run, with a reason — and shown in the dashboard.

Silent cleaning is how data teams lose trust.

A number moves, nobody can explain it, and after that every number gets questioned. So the dashboard carries a permanent line above the metrics: what was excluded, what was deduplicated, and how current the data is.

APPENDIX · HARD TRUTHS

What went wrong, and what it changed

I read past a wrong number.

An early version reported $1.9B at risk across 97 customers — larger than the business. I had multiplied consumption capacity by unit price instead of the commercial quantity. The pipeline reported success throughout.

There is now a test that asserts the portfolio value is plausible. Structure was never the problem; magnitude was.

My own safety mechanism was never exercised.

The audit's first run showed zero rows zero-filled. My generator was emitting zeros where real telemetry emits nothing, so the anti-shelfware logic had never run and its tests were passing vacuously.

The audit caught it, not the tests. That is the argument for having built it.

Synthetic data validates mechanics, not predictive power.

These tests prove the logic handles twelve messy realities. They cannot prove VRR predicts churn.

That needs a backtest against real history — which is exactly what I am asking for.